Syed Huzaifa Ali

Senior Software Engineer | Spatial Computing & Simulation

Trier, Germany (Open to Relocate)

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Professional Summary

Senior Software Engineer with 8 years of experience architecting and delivering production real-time simulations, spatial-computing systems, digital twins, and XR applications across mobile, web, and immersive platforms. Strong in C#, Python, modular software architecture, REST APIs, CI/CD, performance optimization, and real-time networking. Built LiDAR-aware spatial workflows, persistent world mapping, constraint-driven interaction systems, medical simulations, synthetic-data pipelines, and networked avatar experiences. Led a 10-member multidisciplinary team and delivered three enterprise simulation programs on schedule.

Experience

Scientific Assistant – Spatial Computing & Simulation
Fraunhofer-Institute for Solar Energy Systems ISE
Official Project Release: Heat Pump PlanAR
Sep 2024 – Present *Germany*

- Development of WESPE-PlanAR, an iPad-based spatial planning system that enables field engineers to validate heat-pump layouts before physical installation.
- Architected a modular digital-twin workflow with ARKit, spatial constraints, and optimized pipe routing, reducing material costs by 15% and improving spatial accuracy by up to 20%.
- Built persistent spatial mapping with anchors, relocalization, session lifecycle management, versioned world-map storage, and recovery logic, reducing on-site setup time by 30%.
- Engineered an end-to-end 3D data pipeline for GLB import/export, model labeling, LiDAR-mesh collision, and BIM-ready handoff to downstream engineering tools.
- Implemented headless processing to extract room measurements and publish structured outputs without a display context, integrating spatial data into external planning workflows.
- Developed constraint-driven interaction systems for wall pinning, orthogonal rotation, elevation, alternate views, autorouting, pipe management, and multilingual device catalogs.
- Managed provisioning, build validation, and TestFlight/MDM distribution; introduced release, development, bugfix, and deployment branches to improve release traceability.

Technical Lead – Real-Time Software Systems
Digitrends
Jun 2022 – Apr 2024 *USA / Remote*

- Led architecture and delivery for three enterprise real-time 3D and medical-simulation programs for GSK, Abbott, and Roche, achieving 100% on-time release and increasing client retention by 20%.
- Designed modular C# systems, state-management patterns, reusable components, integration boundaries, and performance budgets aligned with product scope and quality targets.
- Built and maintained CI/CD pipelines for automated builds, packaging, and release workflows, improving deployment consistency across concurrent client programs.
- Directed a 10-member engineering, art, QA, and UI/UX team through sprint planning, technical reviews, risk management, and stakeholder reporting.
- Established shared tooling, coding standards, asset pipelines, and analytics-driven iteration practices that reduced rework and increased user engagement by 35%.

Freelance Software Engineer – Interactive Systems
Independent
Jan 2021 – Dec 2023 *Remote*

- Designed an IoT health application architecture with real-time breathing-data visualization, increasing session engagement by 45%.
- Shipped a multiplayer WebGL tournament platform with a custom PHP backend and eBay API integration for authentication, matchmaking, and persistence, increasing participation by 50%.
- Delivered cross-platform AR/WebGL systems and optimized asset loading, LOD, animation, and runtime memory, extending average dwell time by 40%.

Technical Lead – Multiplayer & Web Systems
Imagination Warehouse
Oct 2021 – Jun 2022

- Architected a multiplayer WebGL collaboration platform integrating Photon PUN 2, Agora, and Ready Player Me with scalable room, avatar, and session workflows.
- Integrated real-time networking and voice with permission controls, reconnection safeguards, memory budgets, asset streaming, and failure-recovery flows for reliable distributed sessions.
- Coordinated engineering, 3D, and UI/UX delivery through asset standards and review cadences, increasing user retention by 25%.

Software Engineer – Real-Time Applications
Digimerse
Jun 2020 – May 2021

- Improved load time and rendering performance by 35% through profiling, batching, asset tuning, and runtime optimization.
- Automated build and release workflows with Jenkins, reclaiming 10+ developer hours per week, and documented three core delivery processes to improve maintainability.

Mixed Reality & Simulation Engineer
NED University – CSRD
May 2018 – May 2020 *Karachi, Pakistan*

- Prototyped mobile training, simulation, and HCI systems using ARCore, ARKit, spatial anchoring, plane detection, and raycasting, owning end-to-end implementation and integration.
- Built an immersive Unity/C# VR campus tour for remote exploration of university spaces, owning core interaction and environment integration.

Technical Skills

Programming: C#, Python, JavaScript, PHP, LINQ, HTML
Software Engineering: OOP, SOLID, system architecture, modular design, state management, REST APIs, JSON/XML, CI/CD, build validation, performance profiling, real-time networking, failure recovery
Simulation, Spatial Computing & XR: Unity, real-time simulation, synthetic data generation, domain randomization, object detection, ARFoundation, ARCore, ARKit, OpenXR, Meta XR, XR Interaction Toolkit, LiDAR, digital twins, BIM/GLB workflows
Platforms & Tools: Git/GitHub, Jenkins, Jira, Firebase, App Store Connect, TestFlight, iOS, Android, PC, WebGL
SDKs: Photon PUN 2, Agora, Ready Player Me, VRM/UniVRM, Vuforia, Addressables
Languages: English (C1), German (A1)

Education

M.Sc. Computational Linguistics / Natural Language Processing
University of Trier
Apr 2024 – Apr 2027 *Trier, Germany*

Master of Computer Science
Bahria University
Aug 2019 – Aug 2021 *Karachi, Pakistan*

Bachelor of Computer Science
MAJU University
Jan 2015 – Dec 2018 *Karachi, Pakistan*